

Dockerfile

A dockerfile is a text file that contains a set of instructions to build a Docker image, it tells Docker what base image to use, what software to install, what files to copy, and what command to run when the container starts.

Why use a Dockerfile?

- ∴ Automation - Build images automatically instead of creating them manually.
- Repeatability :- The same Dockerfile will always create the same image.
- Portability :- share Dockerfile with others so they can build the same environment.
- Version Control - store in Git to track changes.
- Lightweight Deployments :- only required components are included

Key Points :-

- written in plain text using Dockerfile syntax.
- Each instruction creates a layer in the image.
- Common instructions: FROM, RUN, COPY, ADD, WORKDIR, ENV, EXPOSE, CMD, ENTRYPOINT.

1. Simple Ubuntu image example

- FROM ubuntu:18.04
- RUN echo "Hello All welcome to Docker" > /tmp/myfile

- FROM → sets the base image (ubuntu:18.04).
- RUN → Executes a command during image build (here, creating /tmp/myfile with text).

2. Docker file with multiple instructions.

- FROM ubuntu:18.04
- WORKDIR /tmp
- RUN echo "Hello All" > myFile
- ENV name shantany
- COPY sample-war /tmp
- ADD shantany.tar.gz /tmp

- WORKDIR → sets working directory inside image
- RUN → creates myFile with content
- ENV → sets environment variable name = shantany.
- COPY → copies sample-war from host to container.
- ADD → Adds shantany.tar.gz and auto-extracts it.

3. installing multiple packages

X Non-optimized;

- FROM ubuntu:18.04
- RUN apt-get update
- RUN apt-get install curl -y
- RUN apt-get install apache2 -y
- RUN apt-get install vim -y

- multiple RUN layers → Larger image

✓ Optimized :-

- FROM ubuntu:18.04
- RUN apt-get update && apt-get install vim curl apache2 -y

- combine into one layer for smaller, faster builds.

4. CMD vs ENTRYPOINT

CMD Example:

- FROM centos:8
- CMD ping localhost -c 20

• CMD → Default command (can be overridden when running the container).

ENTRYPOINT Example:

- FROM centos:8
- ENTRYPOINT ping localhost -c 20

• ENTRYPOINT → main executable that cannot be overridden easily.

5. ENTRYPOINT (Exec form)

- FROM centos:8
- ENTRYPOINT ["ping", "localhost", "-c", "10"]

• Exec form is preferred - avoids shell and works better with arguments.

6. Apache HTTPD with Ubuntu

- FROM ubuntu:18.04
- RUN apt-get update & apt-get install -y apache2
- RUN echo "ServerName localhost" >> /etc/apache2/apache2.conf
- COPY index.html /var/www/html
- EXPOSE 80
- ENTRYPOINT ["/usr/sbin/apachectl", "-D", "FOREGROUND"]

- install apache, sets server name, copies index.html to web root, exposes port 80, and keeps apache running in foreground.

7. Apache HTTPD with Centos

- FROM centos:8
- RUN cd /etc/yum.repos.d/
- RUN sed -i 's/mirrorlist/#mirrorlist/g' /etc/yum.repos.d/centos-*
- RUN sed -i 's|#baseurl=http://mirror.centos.org|baseurl=http://vault.centos.org/g' /etc/yum.repos.d/centos-*
- RUN yum install httpd -y
- COPY index.html /var/www/html/
- EXPOSE 80
- ENTRYPOINT ["/usr/sbin/httpd", "-DFOREGROUND"]

• update centos repo URLs (needed for centos 8 since default mirrors are archived)

- installs Apache (httpd) and serves custom index.html.

Key Dockerfile Instructions Recap

<u>instruction</u>	<u>purpose</u>
FROM	Base image to use
RUN	Executes commands at build time
WORKDIR	Sets default working directory
COPY	copies files from host to image
ADD	copies files (also extracts archives)
ENV	sets environment variables.
EXPOSE	informs Docker of container port
CMD	Default command to run
ENTRYPOINT	main executable of container.